

AMBERIEU-EN-BUGEY AB, France

WMO: 074820

Lat: 45.9765N

Lon: 5.3293E

Elev: 251

StdP: 98.35

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-5.3	-9.9	1.7	-6.3	-7.6	2.0	-3.8	10.4	5.3	9.3	6.2	2.0	10	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	32.9	20.6	30.8	20.2	29.0	19.7	21.8	30.0	21.0	28.8	20.2	27.5	3.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.3	25.1	18.4	13.7	24.0	17.6	13.0	23.1	64.5	30.1	61.6	28.9	59.0	27.7	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	6.9	DB	-10.6	35.3	3.1	2.2	-12.9	36.9	-14.7	38.1	-16.4	39.3	-18.7	40.9	(4)
(5)			WB	-10.9	23.3	3.1	0.8	-13.1	23.9	-14.9	24.4	-16.6	24.8	-18.8	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	3.1	4.1	7.9	11.2	15.2	19.0	21.0	20.5	16.4	12.6	7.0	3.7	(6)	
	DBStd	7.48	4.42	4.36	3.85	3.64	3.59	3.78	3.48	3.54	3.46	4.04	4.44	4.51	(7)	
	HDD10.0	832	216	167	87	29	3	0	0	0	1	21	110	199	(8)	
	HDD18.3	2636	472	397	324	214	109	37	13	17	78	179	341	453	(9)	
	CDD10.0	1510	2	3	22	65	164	271	341	326	192	102	20	4	(10)	
	CDD18.3	272	0	0	0	1	12	57	96	84	19	3	0	0	(11)	
	CDH23.3	2850	0	0	0	19	133	612	1017	881	178	11	0	0	(12)	
	CDH26.7	1013	0	0	0	1	22	203	389	362	36	0	0	0	(13)	
(14) Wind	WSAvg	2.8	2.8	3.0	3.2	3.3	2.9	2.9	2.7	2.5	2.5	2.6	2.8	2.7	(14)	
(15) Precipitation	PrecAvg	1152	89	81	73	94	103	88	87	83	111	121	124	97	(15)	
	PrecMax	1412	176	184	255	208	219	172	229	207	330	253	327	185	(16)	
	PrecMin	814	27	12	12	18	21	23	28	23	28	26	52	14	(17)	
	PrecStd	154	40	49	50	51	47	38	46	41	74	52	65	39	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.6	21.4	26.0	29.1	33.9	35.0	36.5	30.1	24.7	19.2	15.2	(19)	
		MCWB	10.4	9.8	12.6	15.5	18.7	21.3	20.7	20.9	19.1	17.2	13.6	10.9	(20)	
	2%	DB	12.3	14.2	18.9	22.8	26.7	31.0	32.8	33.0	27.5	22.7	17.1	13.3	(21)	
		MCWB	9.2	9.1	11.5	14.3	17.1	20.3	20.6	20.6	18.9	16.6	12.5	10.1	(22)	
	5%	DB	11.1	12.4	16.8	20.7	24.6	29.0	30.7	30.6	25.4	20.8	15.4	12.1	(23)	
		MCWB	8.5	8.3	10.7	13.2	16.5	19.7	20.3	20.1	17.9	15.7	11.9	9.3	(24)	
	10%	DB	9.6	10.8	14.8	18.5	22.6	27.0	28.7	28.2	23.3	19.0	13.7	10.5	(25)	
		MCWB	7.4	7.6	9.8	12.2	15.6	18.7	19.6	19.6	17.2	14.8	11.0	8.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.7	13.3	16.5	19.7	22.9	22.8	22.6	21.1	18.2	14.5	11.7	(27)	
		MCDB	13.2	14.3	19.5	24.4	26.6	30.7	30.9	31.8	26.7	22.7	17.9	13.9	(28)	
	2%	WB	9.7	9.7	12.2	14.9	18.4	21.4	21.7	21.5	19.5	17.2	13.1	10.5	(29)	
		MCDB	11.8	13.1	17.1	21.5	25.0	29.0	30.0	30.2	25.6	21.2	16.1	12.7	(30)	
	5%	WB	8.7	8.8	11.2	13.7	17.1	20.3	20.9	20.7	18.6	16.3	12.2	9.6	(31)	
		MCDB	10.8	11.9	15.8	19.6	22.9	27.5	28.9	28.6	24.2	19.9	14.9	11.7	(32)	
	10%	WB	7.5	7.8	10.2	12.6	16.0	19.3	20.1	19.9	17.5	15.2	11.2	8.5	(33)	
		MCDB	9.4	10.5	14.3	17.7	21.4	25.8	27.5	27.0	22.3	18.5	13.6	10.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	8.1	10.6	11.2	10.9	11.8	12.2	12.2	11.1	9.1	7.1	6.0	(35)	
		MCDBR	7.6	11.5	13.9	15.1	14.0	14.4	15.1	16.1	14.2	11.4	9.4	7.8	(36)	
	5% WB	MCWBR	5.2	7.2	7.8	7.8	6.8	6.2	5.9	6.0	6.7	6.2	5.9	5.3	(37)	
		MCWBR	7.1	10.1	12.4	13.8	12.7	13.7	13.6	14.1	12.4	10.2	8.6	6.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.308	0.329	0.360	0.396	0.403	0.404	0.401	0.394	0.377	0.362	0.329	0.306	(40)	
		taud	2.463	2.406	2.350	2.259	2.289	2.320	2.326	2.354	2.385	2.423	2.469	2.472	(41)	
	Ebn at Noon		798	846	865	861	866	863	861	853	837	794	762	758	(42)	
		Edh at Noon	68	87	106	127	128	125	122	115	102	85	68	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.21	2.06	3.36	4.51	5.26	6.25	6.08	5.26	4.03	2.51	1.38	1.01	(44)	
	RadStd		0.14	0.30	0.42	0.59	0.52	0.52	0.52	0.40	0.31	0.24	0.15	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page